

MPG No. 23.3.84 $6'/m$ [Type III, hexagonal] [M tensor]

* Rank 0 tensor. * Rank 1 tensor. * Rank 2 tensor (s). * Rank 2 tensor (a). * Rank 3 tensor (ss). * Rank 3 tensor (sa). * Rank 3 tensor (a). * Rank 4 tensor (sss).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & M_{xxx} & 0 \\ 0 & 0 & 0 & -M_{xyz} & -M_{xxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ M_{xyz} & -M_{xyz} & 0 & 0 & 0 & -M_{xxx} \\ M_{xxx} & -M_{xxx} & 0 & 0 & 0 & M_{xyz} \\ 0 & 0 & 0 & -M_{xxx} & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = \mathbb{M}_{4a}^{(1)}$$

$$M_{xxx} = \mathbb{M}_{4b}^{(1)}$$

* Rank 4 tensor (ssa).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & M_{xxx} & 0 \\ 0 & 0 & 0 & -M_{xyz} & -M_{xxx} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -M_{xyz} & M_{xyz} & 0 & 0 & 0 & M_{xxx} \\ -M_{xxx} & M_{xxx} & 0 & 0 & 0 & -M_{xyz} \\ 0 & 0 & 0 & -M_{xxx} & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = 2\mathbb{T}_{3a}^{(1)}$$

$$M_{xxx} = -2\mathbb{T}_{3b}^{(1)}$$

* Rank 4 tensor (aas). * Rank 4 tensor (aaa). * Rank 4 tensor (sa).

$$\begin{bmatrix} M_{xyz} & M_{xxx} & 0 \\ -M_{xyz} & -M_{xxx} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ M_{xxx} & -M_{xyz} & 0 \end{bmatrix}$$

$$M_{xyz} = \mathbb{T}_{3a}^{(2)}$$

$$M_{xxx} = \mathbb{T}_{3b}^{(2)}$$

* Rank 4 tensor (as).

$$\begin{bmatrix} M_{yzz} & -M_{yzz} & 0 & 0 & 0 & M_{yzy} \\ M_{yzy} & -M_{yzy} & 0 & 0 & 0 & -M_{yzz} \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzz} = \mathbb{T}_{3a}^{(3)}$$

$$M_{yzy} = \mathbb{T}_{3b}^{(3)}$$

* Rank 4 tensor (s).

$$\begin{bmatrix} 0 & 0 & 0 & M_{xyz} & M_{xxx} & 0 & M_{xzy} & M_{xxz} & 0 \\ 0 & 0 & 0 & -M_{xyz} & -M_{xxx} & 0 & -M_{xzy} & -M_{xxz} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ M_{yzz} & -M_{yzz} & 0 & 0 & 0 & M_{yzy} & 0 & 0 & M_{yzy} \\ -M_{yzy} & M_{yzy} & 0 & 0 & 0 & M_{yzz} & 0 & 0 & M_{yzz} \\ 0 & 0 & 0 & -M_{xxx} & M_{xzy} & 0 & -M_{xxz} & M_{xyz} & 0 \end{bmatrix}$$

$$M_{xxyz} = \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} + \mathbb{T}_{3a}^{(2)}$$

$$M_{xxzx} = \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} + \mathbb{T}_{3b}^{(2)}$$

$$M_{xxzy} = \mathbb{M}_{4a}^{(1)} + 2\mathbb{T}_{3a}^{(1)} - \mathbb{T}_{3a}^{(2)}$$

$$M_{xxzx} = \mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)} - \mathbb{T}_{3b}^{(2)}$$

$$M_{yzxx} = \mathbb{M}_{4a}^{(1)} - 2\mathbb{T}_{3a}^{(1)}$$

$$M_{yzxy} = -\mathbb{M}_{4b}^{(1)} - 2\mathbb{T}_{3b}^{(1)}$$

* Rank 4 tensor (a).

$$\begin{bmatrix} M_{yzxx} & -M_{yzxx} & 0 & 0 & 0 & M_{yzxy} & 0 & 0 & M_{yzxy} \\ M_{yzxy} & -M_{yzxy} & 0 & 0 & 0 & -M_{yzxx} & 0 & 0 & -M_{yzxx} \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$M_{yzxx} = \mathbb{T}_{3a}^{(3)}$$

$$M_{yzxy} = \mathbb{T}_{3b}^{(3)}$$

* Rank 4 tensor (t).

$$\begin{bmatrix} 0 & 0 & M_{xxxx} \\ 0 & 0 & M_{yyyy} \\ 0 & 0 & 0 \\ -M_{xxxx} & M_{yyyy} & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -M_{yyyy} \\ 0 & 0 & 0 \\ M_{xxxx} & -M_{yyyy} & 0 \\ 0 & 0 & -M_{xxxx} \\ -M_{yyyy} & -M_{xxxx} & 0 \end{bmatrix}$$

$$M_{xxxx} = \mathbb{M}_{4b}^{(1)}$$

$$M_{yyyy} = -\mathbb{M}_{4a}^{(1)}$$